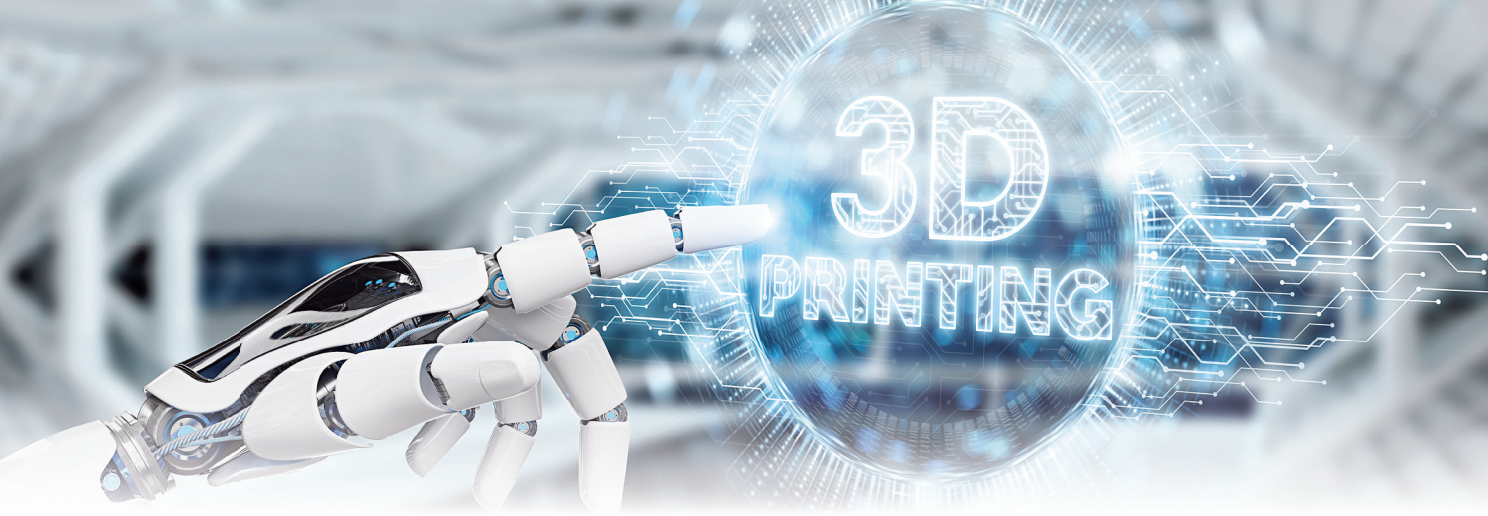


Artificial intelligence takes on many different forms in healthcare. The primary trend for AI in healthcare 2022 will be in utilizing machine learning to evaluate large amounts of patient data and other information. By creating tailored algorithms, programmers can mimic human thought and write programs that can seemingly think, learn, make decisions, and take action. By analyzing healthcare data in this robust and comprehensive way, healthcare leaders will be able to use the findings to improve patient outcomes, reduce costs, and boost staff job satisfaction. AI is also helping clinicians to create more comprehensive treatment programs, allowing patients to manage their conditions more effectively. Neuroscience, the study of the nervous system, encompasses all components that are developed to understand the brain, visualize its processes and even control, repair or improve its functions. It holds major potential for treating brain disorders. Curing cancer is certainly one of the big challenges of the 21st century. Our knowledge of cancer has greatly improved in the last two decades. In recent years, there has been a surge of new technologies aiming to help the immune system identify and attack tumors, a field known as immuno-oncology. These technologies could make a big difference in the way we treat cancer, taking us closer to being able to cure this disease. COVID-19 has impacted the capacity of healthcare systems to diagnose and treat patients with cancer. Precision oncology is the new weapon to defeat cancer. As a foundation and gateway for precision medicine, genetic sequencing can find changes in the human body through the detection of genetic variations and can guide diagnosis and treatment.

Another recent technology, 3D printing uses computer-aided design to create three-dimensional objects through a layering method. 3D printing is being used in the medical sector to help save lives by printing organs for the human body such as livers, kidneys and hearts. Further advances and uses are being developed in the healthcare sector providing some of the biggest advances in using the technology. Blockchain technology can help in addressing the interoperability challenges that health information and technology systems face. The health blockchain will contain a complete indexed history of all medical data, including formal medical records and health data. Data will be sourced from mobile applications and wearable sensors. This will help in seamless medical attention. it allows for storing of data in a secure and authenticated network. Thereby, it will prevent the erosion of Individuals' privacy and ensure data security. Virtual reality has been around for some time. VR allows healthcare professionals to learn new skills and upgrade existing ones in a safer ambiance without any danger to the patients. VR surgery is performed by robotics controlled by human surgeons reducing the time risk of complications. VR is also being used for training and research purposes. Virtual and augmented environments have great potential to change the medical field's environment by making items like remote operations a real possibility. These technologies, such as Post Traumatic Stress Disorder (PTSD), have also been used to treat and cure psychological disorders. In critical situations, augmented reality applications can deliver real-time information to the treatment area to support diagnosis, surgery and treatment plans. AccuVein is a handheld device that can scan the vein network of a patient that leads to a 45% reduction in escalations. Surgeons can plan procedures before making the first cut, models can be made of tumors, and AR diagnostic tools can model disease conditions. Telemedicine is the use of telecommunications and information technology to provide clinical healthcare at a distance. Telemedicine is helpful for people living in rural areas. This is very crucial for India since around 70% of the population lives in rural areas. Telemedicine helps overcome the shortage of health professionals. Telemedicine is a better option while dealing with particularly infectious diseases like COVID and swine flu. Telemedicine is a very useful way for the government to provide affordable healthcare to all, with the help of private players and NGOs. Telemedicine saves time and cost. Further, these platforms are in line with the government's vision of Digital India and are necessary to tackle situations created by a pandemic like COVID-19.

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The technological advancements in medical devices have catalyzed an overall improved treatment and care environment in the country. If planned and implemented well, technological progress can make a significant contribution to India's healthcare needs. Such an investment would impact generations to come with lower healthcare costs and put India on the world map as a country to be reckoned as a thought leader in adopting digital health. India needs a holistic approach to tackle problems in the healthcare industry. This includes the active collaboration of all stakeholders' public, private sectors, and individuals. Modern medical technologies show a glimpse into the future. We need to keep on pushing the limits to make health facilities accessible for all. Finally, healthcare quality can be addressed by improving the transparency of healthcare information, innovative technologies, and their application for the betterment of humankind.